

Name: _____

Date: _____

Class: _____

eMyDesign Portfolio

Element H Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to structure your testing plan, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own testing plan, be sure to delete the instructions and delete the examples given in the template table.

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
 - how the test will be set up
 - what needs to happen during the test
 - test cases (if applicable)
 - number of trials of the test expected (aim for 7 or more trials)
 - the expected range of results that could be recorded, including units
 - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
 - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	organized	1) students are able to find a specific type of battery within 15 seconds 1) Place testing battery charging station that is distinguished by brands on table 2) Place 4 batteries in a random stack adjacent to the testing charging station 3) Tester will face away from batteries and charging station and the time starts once tester turns around 4) Have tester organize batteries based on descriptions as quick as possible 5) Stop time once batteries are organized	a. 5	2-8 seconds	Fail: Pass: Goal:
2	Ease of use	1) students will be able to plug in 2 batteries in less than 10 seconds Tester are able to connect the plug from the peg board to the wall plug within 5 seconds in seeing the plug and	5	3-8 seconds	Fail: Pass: Goal:

		wall plug			
3	Modular	1)board is able to be rearranged within 15 seconds Steps 1) Before the tester step foot to the testing room we will rearrange the peg board before asking the tester to step in the room 2) We will tell them how they are going to put everything in place and ask the tester to step in the room and stand near the table near the entrance door 3)	5	10-15 sec	1st test success 2nd test success 3rd test fail (prototype broke) 4th test success 5th test success 6th test successw

4	Ease of access	1)bottom of peg board is no more than 5 feet from the ground and point of access is no more than 2 feet away 2) Tape measure will be used to to measure both distance and height 3) Tester will stand on the tape on the ground and will reach for the tap on the wall 4) they will conclude whether or not if they can reach(touch) the tape	5	5 feet up 2 feet away	1st test success 2nd test success 3rd test success 4th test success 5th test success
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Feedback From Field Experts (H5)

Guidance to students about this section of Element H.

- a) "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.
- b) You must provide evidence in this section that your testing plan has been reviewed by 2-3 field experts.
- c) Do not incorporate their feedback in the table above. Rather, make a copy of the table in the section above, paste it into the section below, and make all revisions only in the table below. This provides evidence that you incorporated the feedback and shows your testing plan's progression.

Revised Testing Details That Incorporate Field Expert Feedback (H5)